

EXPERIMENT 3

POLYALPHABETIC CIPHER (VIGENÈRE CIPHER)

AIM

To implement the Vigenère (Polyalphabetic) Cipher substitution technique using C.

ALGORITHM

1. Read the plaintext from the user.
2. Read the keyword from the user.
3. Repeat the keyword until its length matches the plaintext.
4. Encrypt each character by adding the positions of the plaintext character and the corresponding key character modulo 26.
5. Display the encrypted (cipher) text.
6. Decrypt the cipher text by subtracting the key character value from the cipher character modulo 26.
7. Display the decrypted text.

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char text[100], key[100];
    int i, len;

    printf("Enter Plain Text: ");
    scanf("%s", text);

    printf("Enter Key: ");
    scanf("%s", key);

    len = strlen(key);

    printf("Encrypted Text: ");
    for(i=0; text[i]!='\0'; i++) {
        printf("%c", ((text[i]-'A'+key[i%len]-'A')%26)+'A');
    }

    return 0;
}
```

OUTPUT

```
Output
Enter Plain Text: active
Enter Key: 3
Encrypted Text: SULANW

=== Code Execution Successful ===
```

RESULT

.Thus, the Vigenère (Polyalphabetic) Cipher substitution technique was implemented successfully using C.